

Logic Coverage Criteria: Applied to test code

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TriTyp program

```
1 import java.io.*;
2 public class TriTyp
{
  3  private static String[] triTypes = { "", // Ignore 0.
  "scalene", "isosceles", "equilateral", "not a valid triangle"};
  4 public static void main (String[] argv)
  {  // Driver program for TriTyp
5    int A, B, C;
6    int T;
7    System.out.println ("Enter sides of a triangle");
8    System.out.println ("Enter side 1: ");
9    A = getN();
10   System.out.println ("Enter side 2: ");
11   B = getN();
12   System.out.println ("Enter side 3: ");
13   C = getN();
14   T = triang (A, B, C);
15   System.out.println ("Result is: " + triTypes[T]);
}
```

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TriTyp program

```
// The main triangle classification method
16 private static int triang (int Side1, int Side2, int Side3)
{
17   int triOut;
   // triOut is output from the routine:
   //    triang = 1 if triangle is scalene
   //    triang = 2 if triangle is isosceles
   //    triang = 3 if triangle is equilateral
   //    triang = 4 if not a triangle

   // After a quick confirmation that it's a valid
   // triangle, detect any sides of equal length
18   if (Side1 <= 0 || Side2 <= 0 || Side3 <= 0)
     {
19     triOut = 4;
20     return (triOut);
     }
```

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TriTyp program

```
21 triOut = 0;
22   if (Side1 == Side2)
23      triOut = triOut + 1;
24   if (Side1 == Side3)
25      triOut = triOut + 2;
26   if (Side2 == Side3)
27      triOut = triOut + 3;
28   if (triOut == 0)
     {// Confirm it's a valid triangle before declaring
      // it to be scalene
29     if (Side1+Side2 <= Side3 || Side2+Side3 <= Side1 ||
        Side1+Side3 <= Side2)
30       triOut = 4;
     else
31       triOut = 1;
32     return (triOut);
   }
```

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TriTyp program

```
   // Confirm it's a valid triangle before declaring
   // it to be isosceles or equilateral

33    if (triOut > 3)
34       triOut = 3;
35    else if (triOut == 1 && Side1+Side2 > Side3)
36       triOut = 2;
37    else if (triOut == 2 && Side1+Side3 > Side2)
38       triOut = 2;
39    else if (triOut == 3 && Side2+Side3 > Side1)
40       triOut = 2;
   else
41       triOut = 4;
42    return (triOut);
} // end Triang
```

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TriTyp program

- The program has three inputs— read into variables `A`, `B` and `C` in the main program method and then passed to the formal parameters `Side1`, `Side2` and `Side3` in `Triang()`.
- Solve for the internal variable `triOut`.
- Solve for reachability of the various predicates and then work with criteria.

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Predicates in TriTyp

- Line 18: (Side1 <= 0 || Side2 <= 0 || Side3 <= 0).
- Line 22: (Side1 == Side2)
- Line 24: (Side2 == Side3)
- Line 26: (Side3 == Side1)
- Line 28: (triOut == 0)
- Line 29: (Side1+Side2 <= Side3 || Side2+Side3 <= Side1 || Side1+Side3 <= Side2)
- Line 33: (triOut > 0)
- Line 35: (triOut == 1 && Side1+Side2 > Side3)
- Line 37: (triOut == 2 && Side1+Side3 > Side2)
- Line 39: (triOut == 3 && Side2+Side3 > Side1)

Reachability for the predicates

- Line 18: True
- Line 22: P1=(S1 > 0 && S2 > 0 && S3 > 0); three sides shortened as S1, S2 and S3
- Lines 24, 26 and 28: P1
- Line 29: P1 && (triOut == 0)
- Else of line 29: P1 && (triOut == 0) &&(S1+S2 > S3) && (S2+S3 > S1) &&(S3+S1 > S2)

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Reachability for the predicates

- Line 33: P1 && (triOut != 0)
- Line 35: P1 && (triOut !=0) && (triOut <= 3)
- Line 37: P1 && (triOut !=0) && (triOut <= 3) &&((triOut !=1) || (S1+S2 <=S3))
- Line 39: P1 && (triOut !=0) && (triOut <= 3) &&((triOut !=1) || (S1+S2 <=S3)) &&((triOut !=2) || (S1+S3 <=S2))
- Else of line 39: P1 && (triOut !=0) && (triOut <= 3) &&((triOut !=1) || (S1+S2 <=S3)) &&((triOut !=2) || (S1+S3 <=S2)) &&((triOut !=3) || (S2+S3 <=S1))

Solving for internal variable `triOut`

`triOut`	Rules for determining `triOut`
0	S1 != S2 && S1 != S3 && S2 != S3
1	S1 == S2 && S1 != S3 && S2 != S3
2	S1 != S2 && S1 == S3 && S2 != S3
3	S1 != S2 && S1 != S3 && S2 == S3
4	S1 == S2 && S1 != S3 && S2 == S3
5	S1 != S2 && S1 == S3 && S2 == S3
6	S1 == S2 && S1 != S3 && S2 == S3

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Reachability for predicates, after solving for `triOut`

- Line 29: P1 && (S1 != S2 && S1 != S3 && S2 != S3)
- Else of line 29: P1 && (S1 != S2 && S1 != S3 && S2 != S3) &&(S1+S2 > S3) && (S2+S3 > S1) &&(S3+S1 > S2)
- Line 33: P1 && (S1 == S2 && S1 == S3 && S2 == S3)
- And so on... keep substituting for `triOut` predicates.

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Predicate coverage for the predicates

		True				False			
	Predicate	A	B	C	EO	A	B	C	EO
line 18	$S1 \leq 0 \vee S2 \leq 0 \vee S3 \leq 0$	0	0	0	4	1	1	1	3
line 22	$(S1 == S2)$	1	1	1	3	1	2	2	3
line 24	$(S1 == S3)$	1	1	1	3	1	2	2	2
line 29	$(S1+S2 \leq S3 \vee$ $S2+S3 \leq S1 \vee$ $S1+S3 \leq S2) \vee$	1	2	3	4	2	3	4	1

Similarly, test cases can be written for other predicates too.

Clause coverage for the predicates

Clause coverage is considered only for the predicates having more than one clause.

		True				False			
	Clause	A	B	C	EO	A	B	C	EO
line 18	(S1 $\leq$ 0)	0	1	1	4	1	1	1	3
	(S2 $\leq$ 0)	1	0	1	4	1	1	1	3
	(S3 $\leq$ 0)	1	1	0	4	1	1	1	3
line 29	(S1+S2 $\leq$ S3)	2	3	6	4	2	3	4	1
	(S2+S3 $\leq$ S1)	6	2	3	4	2	3	4	1
	(S1+S3 $\leq$ S2)	2	6	3	4	2	3	4	1

Similarly, test cases can be written for clause coverage for other predicates too.

CACC for the predicates

	Predicate	Clauses			A	B	C	EO
line 18	S1 $\leq$ 0$\vee$ S2 $\leq$ 0$\vee$ S3 $\leq$ 0	T	f	f	0	1	1	4
		F	f	f	1	1	1	3
		f	T	f	1	0	1	4
line 29	(S1+S2 $\leq$ S3 $\vee$	T	f	f	2	3	6	4
	S2+S3 $\leq$ S1 $\vee$	F	f	f	2	3	4	1
	S1+S3 $\leq$ S2) $\vee$	f	T	f	6	2	3	4
		f	f	T	2	6	3	4

Similarly, test cases can be written for CACC for other predicates too.

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Credits

The program `TriTyp` and the coverage criteria details are from the first edition of the book Introduction to Software Testing, by Paul Ammann and Jeff Offutt.

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